

Advanced (Carolina Reaper)

- Q1.** A bookshelf has 48 books. If they are packed in boxes that hold 8 books each, how many boxes are needed with 0 left-over books?
(A) 5
(B) 6
(C) 7
(D) 8
(E) None of the above
- Q2.** What is the remainder when 27 is divided by 9 using the division algorithm?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 6
- Q3.** If a bakery sells 240 loaves of bread per day, and they are packed 12 loaves per box, how many boxes are sold per day?
(A) 18
(B) 20
(C) 22
(D) 24
(E) 30
- Q4.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces will each friend get if they use the division algorithm?
(A) 4
(B) 5
(C) 6
(D) 7
- (E) 8
- Q5.** What is the remainder when 75 is divided by 5?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q6.** A water tank can hold 180 gallons of water. If 12 gallons of water are added to it every hour, how many hours will it take to fill the tank if it is currently empty?
(A) 12
(B) 15
(C) 18
(D) 20
(E) 25
- Q7.** What is the result of $24 \div 3$ using the division algorithm?
(A) 6 R 0
(B) 7 R 1
(C) 8 R 0
(D) 9 R 1
(E) 10 R 0
- Q8.** A person has 120 minutes to complete a task. If the task requires 8 minutes of work and 2 minutes of rest per cycle, how many complete cycles can the person complete?
(A) 10
(B) 12
(C) 14
(D) 15
(E) 18

Open Ended Questions (FRQ)

- Q9.** In a game, a player scores 3 points for every 5 minutes played. If the player played for 35 minutes, how many points will the player score? Show all your work and express your answer as a division result with remainder.
- Q10.** A library has 288 books to put on shelves that can hold 16 books each. How many shelves will the library need, and how many books will be left over? Use the division algorithm to solve the problem and show all your work.

Chef's Tips

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- Tip 1: Encourage students to use real-life examples to understand the division algorithm.
 - Tip 2: Remind students to check their work by multiplying the quotient by the divisor and adding the remainder.
 - Tip 3: Suggest that students use visual aids like diagrams or charts to help with long division problems.